

Biology Mock 2 — Units 2 and 3

MTOH • 90 MINUTES • 80 MARKS

NAME

DATE

SCORE / 80

- Answer every question in the space provided. Show your working and your reasoning — method marks are awarded even when the final answer is wrong.
- A calculator is allowed, but every number on this paper is designed to work without one.
- The mark for each question is shown in square brackets on the right. There are 80 marks in 90 minutes.
- Where a question says explain, a feature on its own is not enough — say what it achieves. “The wall is thin” is a description; “thin, so the diffusion distance is short” is an explanation.
- Where a question asks for a sequence, write it in order and make each step the cause of the next. Naming only the first and last steps scores the marks for neither.
- Section 1 is recall, Section 2 is explanation, Section 3 is data and problems. Work in order and do not spend more than six minutes stuck on any one question.

SECTION 1 – THE MECHANICS

1. The trachea.

- (a)** Name the tissue that forms rings around the trachea. [1]

- (b)** Explain why these rings are needed. [1]

2. Two types of cell line the airways.

- (a)** Name the cells that produce mucus, and state what the mucus does. [1]

- (b)** State what the cilia do. [1]

3. Aerobic respiration.

(a) Write the word equation for aerobic respiration. [1]

(b) State where in a cell it takes place. [1]

4. Blood on its way to the lungs.

(a) Name the blood vessel that carries blood from the heart to the lungs. [1]

(b) Name the chamber of the heart that this blood leaves from. [1]

5. State two ways in which a red blood cell is adapted for carrying oxygen. [2]

6. Define each of these ecological terms.

(a) Population [1]

(b) Ecosystem [1]

7. Food chains.

(a) State what the arrows in a food chain represent. [1]

(b) State what a producer does that no consumer can do. [1]

8. Energy transfer between trophic levels.

(a) State approximately what percentage of the energy is passed on to the next level. [1]

(b) Name the route by which most of the rest is lost. [1]

9. Fertiliser and rivers.

(a) Name the process by which nitrates are washed from fields into a river. [1]

(b) Name the organisms whose rapid growth follows. [1]

10. Name two human activities that reduce biodiversity. [2]

11. Breathing and respiration.

(a) State one difference between breathing and respiration. [1]

(b) State where in the body respiration takes place. [1]

- 12.** Describe, in order, what happens in the thorax during quiet exhalation. Begin with the muscles and end with the movement of air. [3]

One word earns a mark here and most candidates avoid it because it sounds like doing nothing. The muscles relax.

- 13.** Explain three separate ways in which smoking damages the gas exchange and transport systems. [3]

- 14.** Explain why veins contain valves along their length but arteries do not. [3]

- 15.** Explain what is meant by a double circulation, and give one advantage of having one. [3]

- 16.** Describe three features of a capillary that make it suited to exchanging substances with body cells. In each case state what the feature achieves. [3]

- 17.** Describe what happens to blood pressure between the aorta and the vena cava, and explain how blood is returned to the heart despite this. [3]

- 18.** Explain why a pyramid of numbers is sometimes inverted, while a pyramid of biomass for the same food chain is not. [3]

- 19.** Describe the role of decomposers in an ecosystem, and explain what would happen to the ecosystem without them. [3]

- 20.** Describe how a transect is used, and explain when it is a better method than placing quadrats at random. [3]

- 21.** Explain how the enhanced greenhouse effect causes sea levels to rise, and state one consequence for living organisms. [3]

- 22.** Describe three measures that would reduce eutrophication in a river running past farmland. [3]

SECTION 3 – DATA AND PROBLEMS

- 23.** A student measures his breathing before and after exercise.
At rest: 15 breaths per minute, 500 cm³ per breath. After exercise: 28 breaths per minute, 1750 cm³ per breath.

- (a)** Calculate his ventilation rate at rest and after exercise, both in dm³ per minute. [2]

(b) Calculate how many times greater the second value is, to one decimal place. [1]

(c) Explain why both the rate and the depth of his breathing increased. [2]

24. Fully working blood carries 200 cm^3 of oxygen per dm^3 . A heavy smoker has 15% of his haemoglobin permanently bound to carbon monoxide. His cardiac output at rest is 5.2 dm^3 per minute.

(a) Calculate the volume of oxygen 1 dm^3 of his blood can carry. [1]

(b) Calculate the volume of oxygen his blood delivers each minute, and how much less this is than a non-smoker with the same cardiac output. [2]

(c) His alveoli are undamaged and his breathing rate is normal, yet he is breathless walking uphill. Explain why. [2]

If 15% is out of action, multiply by what is left. Subtracting 15 from 200 is the error this question is built to catch.

25. A student estimates the number of plantains in a field of area 250 m^2 . She places a quadrat measuring $0.25 \text{ m} \times 0.25 \text{ m}$ at eight random positions and counts:
3, 5, 2, 6, 4, 3, 5, 4

- (a)** Calculate the mean number per quadrat and estimate the total number of plantains in the field. [2]

- (b)** Give one way she could make her estimate more reliable. [1]

- (c)** In a predator-prey cycle, the peak in the predator population always comes after the peak in the prey population. Explain why. [2]

The quadrat is a quarter of a metre on each side, so its area is not 0.25 m^2 . Work the area out before you divide by it.

26. A forest covers 480 000 hectares and absorbs about 5.5 tonnes of carbon dioxide per hectare each year. 12 000 hectares are cleared every year.

- (a)** Calculate the mass of carbon dioxide the whole forest absorbs in one year. [2]
Give your answer in tonnes, in standard form.

- (b)** Calculate the reduction in the forest's annual absorption caused by one year of clearing. [1]

- (c)** Explain why the total effect on atmospheric carbon dioxide is greater than your answer to (b). [2]

27. Three answers from another student's mock paper are shown below. Each one is wrong. For each, explain what is wrong with it and give the correct version.

- (a)** "Arteries carry oxygenated blood and veins carry deoxygenated blood." [2]

- (b)** "Fertiliser running into a river poisons the fish and they die." [2]

- (c)** "Energy is recycled around an ecosystem by the decomposers." [1]

A mark for saying what is wrong, a mark for the correction. Naming the rule the student broke is worth more than simply writing a better sentence.

End of paper. Go back to every question that asked you to explain and check that each feature you named is followed by what it achieves. That is where this paper is won and lost.